

IIIT-H
Electronics Workshop-II
Spring-2026

Exam: Project-1 Written Exam

Marks: 20

Date: 10-March-2026

Time: 10:15 to 11:00am

Instructions:

- Answering all the questions is compulsory.
- All steps should be justified in detail.
- Clearly state the assumptions (*if any*) made that are not specified in the questions.

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1. (10 marks) Design a differential amplifier of gain 10 when the supply voltage is ⁺⁵10 volts. Assume transistor current gain to be equal to 100 and collector current to be 1mA. Describe AC and DC analysis and the corresponding steps in detail.
 2. (7 marks) Explain why a multi-stage architecture (pre-amplifier, voltage gain stage, band-pass filter, and power amplifier) is preferred over a single-stage design for amplifying weak audio signals from a microphone to drive a speaker. In your answer, discuss the specific role each stage plays and how impedance matching between stages ensures efficient signal transfer without loading effects.
 3. (1 mark) The pre-amplifier stage uses a differential amplifier configuration primarily because it provides a high Common Mode Rejection Ratio (CMRR). What does a high CMRR ensure?
 - (a) It maximizes the overall voltage gain of the amplifier system.
 - (b) It suppresses noise and interference that appear equally on both input terminals while amplifying the desired differential signal.
 - (c) It reduces the output impedance of the pre-amplifier to zero.
 - (d) It eliminates the need for a band-pass filter in subsequent stages.
 4. (1mark) In the voltage gain stage, a common-emitter amplifier with a bypass capacitor across the emitter resistor is used. What is the primary purpose of this bypass capacitor?
 - (a) To block the DC component from reaching the next stage.
 - (b) To filter out frequencies above 20 kHz at the output.
 - (c) To provide a low-impedance path for AC signals, effectively increasing the voltage gain at signal frequencies.
 - (d) To stabilize the DC operating point of the transistor.
 5. (1 mark) A Class-AB power amplifier is chosen over a Class-A amplifier for the output stage. Which of the following correctly describes the advantage of Class-AB operation?
 - (a) Class-AB has a conduction angle of exactly 180°, which eliminates all forms of distortion.
 - (b) Class-AB biases both transistors slightly above cutoff to eliminate crossover distortion while maintaining higher efficiency than Class-A.
 - (c) Class-AB uses only a single transistor, making it simpler and cheaper than Class-A.
 - (d) Class-AB achieves 100
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